Finance Research Letters 98 (2026) 109884



Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl





# The impact of AI washing on enterprises' access to bank loans: From the perspective of external governance

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ARTICLE INFO

*Keywords:*
Ai washing
Bank credit
Information disclosure
Financial regulation

ABSTRACT

AI washing refers to an opportunistic practice whereby enterprises intentionally exaggerate or fabricate the application level of their artificial intelligence technologies in information disclosure to attract market attention and seize commercial interests. This study characterizes listed companies' AI washing behavior from the perspectives of "words" (information disclosure) and "deeds" (actual practice), and finds that such behavior facilitates enterprises' access to bank loans, with a more pronounced effect among state-owned enterprises and non-high-tech enterprises. Furthermore, external governance mechanisms can effectively inhibit the promoting effect of AI washing on enterprises' access to bank loans. The research findings provide important implications for commercial banks to optimize credit resource allocation for AI-related enterprises and for government authorities to strengthen the supervision of tech-focused credit.

## 1. Introduction

In the digital economy era, artificial intelligence (AI)—a general-purpose technology that simulates human thought and cognitive processes—has fundamentally reshaped enterprises' strategic decision-making with its strong industrial scalability and capital attractiveness (Aghion et al., 2019). To capitalize on digital economy opportunities, enterprises are racing to adopt AI technologies. As a primary channel for external financing, bank loans have emerged as a critical funding source for enterprises to advance technological commercialization, drive performance growth, and pursue strategic expansion. Yet the market's AI frenzy has spurred industrial growth while also creating opportunities for speculative behavior (Basele et al., 2025). Some enterprises send misleading signals to the credit market via AI washing—the intentional exaggeration or fabrication of their AI technology adoption levels. This opportunistic practice not only erodes the fairness and legitimacy of enterprises' access to bank loans but also impairs the efficiency of credit resource allocation. In emerging markets with underdeveloped financial systems and acute information asymmetry, such speculative behavior poses a notable threat to credit market stability and thus warrants heightened vigilance (Hui et al., 2024).

Existing studies have extensively examined the impact of corporate information disclosure quality on enterprises' access to bank loans (Chen et al., 2019; Wang, 2023), yet the mechanisms and mitigation pathways of false AI-related information disclosure remain underexplored. Signaling theory offers a valuable theoretical framework for addressing this issue: Enterprises engaging in AI washing leverage social reputation to lower banks' perception of credit risk and boost their likelihood of loan approval; however, signal

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https://doi.org/10.1016/j.frl.2026.109884
Received 14 December 2025; Received in revised form 14 March 2026; Accepted 25 March 2026
Available online 25 March 2026
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effectiveness is inherently constrained by the signaling context. External governance mechanisms—including analyst coverage, external audit quality, and financial regulation—can consistently optimize the signaling context and strengthen banks' capacity to detect false reputations. Notably, while existing studies have affirmed the beneficial role of financial technology in mitigating information asymmetry between banks and enterprises (Feng et al., 2025), financial technology struggles to fundamentally identify AI washing owing to the difficulties of verifying large volumes of information and persistent data barriers (Theodorakopoulos et al., 2025). External governance mechanisms may thus effectively address this shortcoming and serve as a pivotal mechanism for curbing such practices. Against this backdrop, this study addresses three core research questions: Does AI washing help enterprises obtain bank loans? Can external governance mechanisms inhibit this effect? Does this effect exhibit heterogeneous characteristics across enterprises with different ownership types and technological attributes?

Using Chinese A-share listed companies from 2010 to 2023 as the research sample, this study innovatively characterizes enterprises' AI washing behavior from the dual dimensions of "words (information disclosure)" and "deeds (actual practice)". It systematically examines the impact of AI washing on enterprises' access to bank loans, tests the inhibitory effect of external governance mechanisms from three dimensions (analyst coverage, external audit quality, and regional financial regulatory intensity), and further analyzes the heterogeneous characteristics arising from ownership type and technological attributes. This research not only enriches the literature on AI washing and corporate credit in the field of corporate finance but also provides targeted empirical evidence for commercial banks to optimize their credit evaluation systems and for government authorities to improve financial regulatory frameworks.

## 2. Theoretical analysis and research hypotheses

The widespread information asymmetry between banks and enterprises is prone to triggering moral hazard and adverse selection in the credit market. As the core business of banks, lending decisions depend heavily on banks' acquisition and processing of various types of information about debtors (Buchak et al., 2018; Luo, 2025). As a key technology driving the transformation of enterprises' value creation methods, AI application level has become a core factor in bank credit considerations (Xing et al., 2026), and enterprises' AI performance has naturally become an important signal to convey their development potential to banks. From the perspective of signaling theory, AI washing is an opportunistic behavior where enterprises deliberately construct and transmit false signals by exploiting information asymmetry between banks and enterprises (Ozturkcan and Bozdağ, 2025) . By exaggerating or fabricating AI technology application levels, enterprises shape a false reputation of "technological leadership" and release misleading signals to the credit market (Stevens et al., 2015). In the absence of effective external constraints, banks struggle to accurately identify such false signals. The misleading information conveyed aligns with commercial banks' lending preferences for AI-related enterprises under the policy orientation of technology finance, reducing banks' subjective perception of corporate credit risk and thereby helping enterprises obtain bank credit resources. Based on the above analysis, this study proposes the following research hypothesis:

**H1:** AI washing helps enterprises obtain bank loans.

Although the development of financial technology has alleviated information asymmetry between banks and enterprises to a certain extent and played a positive role in optimizing corporate debt financing (Guo et al., 2023), some enterprises still face financing difficulties. The core reason lies in the "islandization" and "fragmentation" of financial technology (Sharma et al., 2025). Data barriers between banks and enterprises reduce data processing efficiency—either the information accumulated by enterprises cannot be effectively processed by banks, or banks struggle to conduct accurate risk assessments due to insufficient digital footprints of enterprises, ultimately leaving room for enterprises to engage in AI washing (Song et al., 2026). According to signaling theory, the effectiveness of signals depends not only on the signals themselves but also on the intensity of constraints in the transmission environment.

The involvement of external governance can effectively optimize the signal transmission environment and break this "information cocoon" (Qian et al., 2025). When enterprises face professional and robust external supervision, the concealment of their AI washing strategies will be significantly reduced, and banks' ability to identify exaggerated information disclosure will be correspondingly enhanced, thereby inhibiting the effect of such strategies on obtaining credit support (Shi et al., 2017). Based on the multi-dimensional nature of external governance, this supervisory effect can exert a progressive role in signal environment governance through three levels: As information intermediaries in the capital market, analysts' continuous tracking of enterprises and output of research reports can effectively make up for banks' information disadvantages (Huang et al., 2018); as independent third-party institutions, external auditors are increasingly focusing on verifying the authenticity of AI adoption through professional audit procedures, greatly increasing the probability of detecting AI washing (Qader and Cek, 2023); as a mandatory institutional constraint, regional financial regulatory intensity clarifies information disclosure rules and raises the cost of false disclosure (Ahn et al., 2025). On one hand, it directly inhibits enterprises' motivation to engage in AI washing, and on the other hand, it forces banks to strengthen their responsibility for verifying information on enterprises' AI adoption, reducing the misallocation of credit resources caused by information misjudgment (Liang et al., 2025). Based on the above analysis, this study proposes the following research hypothesis:

**H2:** The more sound the external governance mechanisms, the weaker the promoting effect of AI washing on enterprises' access to bank loans.

## 3. Research design

### 3.1. Sample selection and data sources

This study takes non-financial A-share listed companies in China from 2010 to 2023 as the initial research sample. Samples labeled

ST, delisted during the period, and those with missing data are excluded. Continuous variables are winsorized at the 1st and 99th percentiles to mitigate the impact of outliers. The final sample consists of 20,140 observations. Annual report texts are obtained from the Juchao Information Network (CNINFO) website, while other financial data are sourced from the China Stock Market & Accounting Research Database (CSMAR).

### 3.2. Variables and definitions

**New bank loans.** Measured as the increase in bank loans obtained by enterprises, standardized by total assets.

**AI washing.** AI washing refers to enterprises' strategy of "talking more but doing less" in information disclosure. Therefore, this study identifies enterprises' AI adoption performance from two dimensions—"words" (information disclosure) and "deeds" (actual practice)—and uses the consistency between words and deeds as a proxy indicator for corporate AI washing. The specific method is as follows.

First, measure the "words" of enterprises' AI adoption. Use Python to segment the annual reports of listed companies and count the frequency of AI-related keywords. To reduce the impact of industry heterogeneity, refer to the research of (Loughran and Mcdonald, 2016), take the same year and industry as the basic statistical unit, and calculate the weight coefficient of each keyword (see Formula 1). Sum the weighted word frequencies of all keywords and construct a dummy variable (*TM*). If the value of enterprise *i* is higher than the mean, it is defined as "talking more" ($TM = 1$); otherwise, it is "talking less" ($TM = 0$).

$$w_{c,i} = \begin{cases} \dfrac{1+\ln\left(tf_{c,i}\right)}{1+\ln(l_i)} \ln\dfrac{N}{df_c}, if\, tf_{c,i} > 0 \\ 0, if\, tf_{c,i} \le 0 \end{cases} \tag{1}$$

Where $N$ represents the number of annual report samples of companies in the same industry and year; $df_c$ denotes the number of corporate annual reports containing keyword $c$; $tf_{c,i}$ indicates the original frequency of keyword $c$ in the annual report of enterprise $i$; and $l_i$ stands for the text length of the annual report of enterprise $i$.

Second, measure the "deeds" of enterprises' AI adoption. Use the AI investment level as a proxy variable, and construct a dummy variable (*DL*). If the AI investment level of enterprise $i$ is lower than the industry-year mean, it is defined as "doing less" ($DL = 1$); otherwise, it is "doing more" ($DL = 0$).

Finally, $AIwashing = 1$ if an enterprise simultaneously exhibits "talking more" and "doing less" ($TM = 1$ and $DL = 1$); otherwise, $AIwashing = 0$.

To minimize the impact of potential confounding variables, a series of control variables are introduced Table 1.

### 3.3. Model construction

Our baseline model is as follows:

$$Loan_{i,t} = \beta_0 + \beta_1 AIwashing_{i,t-1} + \beta_2 Controls_{i,t-1} + \mu_i + \sigma_{t-1} + \varepsilon_{i,t} \tag{2}$$

Where $i$ and $t$ respectively denote firm and year, *Loan* represents the new bank loans, *AIwashing* stands for the company's "talking more but doing less" in AI adoption, *Controls* denotes the set of control variables, and $\beta_1$ is the regression coefficient of the core explanatory variable.

**Table 1**
Variable definitions.

| Variable meaning | Abbreviation | Calculation method |
|---|---|---|
| New Bank Loans | *Loan* | (Total bank loans in the current year - Total bank loans in the previous year) / Total assets at the beginning of the year |
| AI Washing | *AIwashing* | 1 if the enterprise exhibits "talking more but doing less" in AI adoption; otherwise 0 |
| Firm Age | *Age* | Natural logarithm of (current year - establishment year + 1) |
| Financial Leverage | *Lev* | (Net Income + Tax Expense + Interest Expense) / (Net Income + Tax Expense) |
| Return on Equity | *ROE* | Net income divided by shareholders' equity |
| SA index | *SA* | SA index (Hadlock and Pierce, 2010) |
| Tobin's Q | *TobinQ* | Market value to total assets |
| Firm Size | *Size* | Natural logarithm of the number of employees |
| Fixed Asset Ratio | *Fixed* | Net fixed assets to total assets |
| Operating revenue growth | *Growth* | (Operating revenue in current year - Operating revenue in previous year) / Operating revenue in previous year |
| Ownership concentration | *Top*10 | Shareholding percentage of the top ten shareholders |
| CEO—Chairman Duality | *Dual* | 1 if the chairman and general manager are the same person; otherwise 0 |

## 4. Empirical results

### 4.1. Descriptive statistics

Table 2 presents the descriptive statistics for the main variables.

### 4.2. Baseline results

The progressive regression results in Table 3 show that the coefficient of *AIwashing* is significantly positive at the 1 % level (0.0061–0.0087), indicating that AI washing helps enterprises obtain bank loans, which supports H1.

### 4.3. Robustness tests

Alternative Dependent Variable. Incorporate long-term liabilities due within one year into total liabilities to construct a new indicator (*Loan*1). The regression result in Column (1) of Table 4 shows that AI washing systematically increases enterprises' new loans.

Adding Fixed Effects. Control firm-year-region and firm-year-industry fixed effects respectively. Columns (2) and (3) of Table 4 show that the robustness of the benchmark result.

Adjusting the Sample Scope. Exclude data from 2015 (stock market crash) and 2019 (COVID-19 pandemic). As shown in the Column (4) of Table 4, our conclusion still holds.

### 4.4. Mechanisms analysis

Our theoretical analysis in Section 2 indicates that the more robust the external governance mechanisms, the weaker the promoting effect of AI washing on enterprises' access to bank loans. To test this mechanism, we perform an empirical analysis from three perspectives: analyst coverage, external audit quality, and the intensity of regional financial regulation. The results in Table 5 show that the interaction terms between AI washing and analyst coverage, external audit quality, and regional financial regulatory intensity are negative and significant at least at the 10 % statistical level, indicating that external governance mechanisms significantly inhibit the positive effect of AI washing on enterprises' bank credit scale. These results confirm H2.

### 4.5. Heterogeneity analysis

Table 6 shows that AI washing significantly increases the bank credit scale of state-owned enterprises and non-high-tech enterprises.

The heterogeneous response between state-owned and non-state-owned enterprises stems from institutional differences. State-owned enterprises are subject to strong policy intervention and linked to government credit, leading banks to have lower risk sensitivity and looser credit review standards for them. In contrast, banks adopt stricter review standards and conduct more detailed pre-loan due diligence for non-state-owned enterprises, making it difficult for them to benefit from AI washing.

The heterogeneous response between high-tech and non-high-tech enterprises originates from industry differences. The reason may be that banks regard AI application as a routine business activity for high-tech enterprises, so the effect of AI washing is limited. In contrast, AI-related disclosure by non-high-tech enterprises is more eye-catching, enabling AI washing to better attract banks' attention and align with their lending orientation.

## 5. Conclusion

This study finds that AI washing exerts a significant distorting effect on enterprises' access to loans, while external governance

**Table 2**
Descriptive statistics.

| Variables | N | Mean | StdDev | Min | Max |
|---|---|---|---|---|---|
| *Loan* | 20,140 | 0.0268 | 0.0954 | −0.1867 | 0.4647 |
| *AIwashing* | 20,140 | 0.3777 | 0.4848 | 0 | 1 |
| *Age* | 20,140 | 2.9569 | 0.3291 | 1.9459 | 3.6376 |
| *Lev* | 20,140 | 1.4462 | 1.3686 | −1.6674 | 9.9108 |
| *ROE* | 20,140 | 0.0476 | 0.1538 | −0.9002 | 0.3393 |
| *SA* | 20,140 | −3.8380 | 0.2770 | −4.5464 | −2.9826 |
| *TobinQ* | 20,140 | 1.8326 | 1.1676 | 0.8164 | 7.9120 |
| *Size* | 20,140 | 8.0471 | 1.2983 | 4.6052 | 11.4764 |
| *Fixed* | 20,140 | 0.2406 | 0.1724 | 0.0020 | 0.7193 |
| *Growth* | 20,140 | 0.1537 | 0.3983 | −0.5587 | 2.5375 |
| *Top*10 | 20,140 | 0.5674 | 0.1590 | 0.2197 | 0.9115 |
| *Dual* | 20,140 | 0.2160 | 0.4116 | 0 | 1 |

**Table 3**
Baseline regression.

| Variables | (1)<br>*Loan* | (2)<br>*Loan* | (3)<br>*Loan* |
|---|---|---|---|
| *AIwashing* | 0.0086*** | 0.0061*** | 0.0063*** |
| | (0.0024) | (0.0023) | (0.0023) |
| *Age* | | −0.0220 | −0.0071 |
| | | (0.0163) | (0.0172) |
| *Lev* | | −0.0019*** | −0.0018*** |
| | | (0.0007) | (0.0007) |
| *ROE* | | 0.0099** | 0.0096** |
| | | (0.0046) | (0.0045) |
| *SA* | | −0.0804*** | −0.0860*** |
| | | (0.0183) | (0.0185) |
| *TobinQ* | | 0.0153*** | 0.0157*** |
| | | (0.0016) | (0.0016) |
| *Size* | | −0.0114*** | −0.0118*** |
| | | (0.0025) | (0.0025) |
| *Fixed* | | −0.0848*** | −0.0814*** |
| | | (0.0150) | (0.0154) |
| *Growth* | | 0.0057** | 0.0046* |
| | | (0.0025) | (0.0025) |
| *Top*10 | | | 0.0368** |
| | | | (0.0143) |
| *Dual* | | | 0.0027 |
| | | | (0.0031) |
| *Cons* | 0.0391*** | −0.1198 | −0.1997** |
| | (0.0029) | (0.0807) | (0.0876) |
| *Firm* | Yes | Yes | Yes |
| *Year* | Yes | Yes | Yes |
| N | 15,968 | 15,968 | 15,968 |
| adj. $R^2$ | 0.0172 | 0.0418 | 0.0426 |

Notes: Robust standard errors clustered at the firm level are in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

**Table 4**
Robustness tests.

| Variables | (1)<br>*Loan*1 | (2)<br>*Loan* | (3)<br>*Loan* | (4)<br>*Loan* |
|---|---|---|---|---|
| *AIwashing* | 0.0061** | 0.0066*** | 0.0063*** | 0.0108*** |
| | (0.0025) | (0.0023) | (0.0023) | (0.0027) |
| *Age* | −0.0017 | −0.0116 | −0.0071 | 0.0002 |
| | (0.0194) | (0.0175) | (0.0172) | (0.0187) |
| *Lev* | −0.0020*** | −0.0018*** | −0.0018*** | −0.0027*** |
| | (0.0007) | (0.0007) | (0.0007) | (0.0009) |
| *ROE* | 0.0098** | 0.0094** | 0.0096** | 0.0113** |
| | (0.0042) | (0.0046) | (0.0045) | (0.0049) |
| *SA* | −0.1077*** | −0.0886*** | −0.0860*** | −0.0757*** |
| | (0.0206) | (0.0185) | (0.0185) | (0.0182) |
| *TobinQ* | 0.0178*** | 0.0152*** | 0.0157*** | 0.0125*** |
| | (0.0019) | (0.0015) | (0.0016) | (0.0018) |
| *Size* | −0.0100*** | −0.0125*** | −0.0118*** | −0.0080*** |
| | (0.0028) | (0.0026) | (0.0025) | (0.0026) |
| *Fixed* | −0.0945*** | −0.0827*** | −0.0814*** | −0.0729*** |
| | (0.0170) | (0.0156) | (0.0154) | (0.0170) |
| *Growth* | 0.0035 | 0.0042* | 0.0046* | 0.0029 |
| | (0.0026) | (0.0025) | (0.0025) | (0.0030) |
| *Top*10 | 0.0259* | 0.0337** | 0.0368** | 0.0439*** |
| | (0.0155) | (0.0145) | (0.0143) | (0.0164) |
| *Dual* | 0.0011 | 0.0025 | 0.0027 | 0.0029 |
| | (0.0034) | (0.0031) | (0.0031) | (0.0036) |
| *Cons* | −0.2870*** | −0.1924** | −0.1997** | −0.2111** |
| | (0.0982) | (0.0898) | (0.0876) | (0.0895) |
| *Firm* | Yes | Yes | Yes | Yes |
| *Year* | Yes | Yes | Yes | Yes |
| *Region* | No | Yes | No | No |
| *Industry* | No | No | Yes | No |
| N | 15,291 | 15,953 | 15,968 | 11,296 |
| adj. $R^2$ | 0.0457 | 0.0472 | 0.0426 | 0.0423 |

**Table 5**
Mechanism analysis (external governance).

| Variables | (1)<br>Loan | (2)<br>Loan | (3)<br>Loan |
|---|---|---|---|
| *AIwashing* | 0.0081*** | 0.0084*** | 0.0079*** |
| | (0.0027) | (0.0030) | (0.0024) |
| *AIwashing* × *Analyst* | −0.0013* | | |
| | (0.0007) | | |
| *Analyst* | 0.0013** | | |
| | (0.0006) | | |
| *AIwashing* × *Audit* | | −0.0018** | |
| | | (0.0009) | |
| *Audit* | | −0.0045*** | |
| | | (0.0011) | |
| *AIwashing* × *Regulatory* | | | −0.1910** |
| | | | (0.0964) |
| *Regulatory* | | | 0.0194 |
| | | | (0.0516) |
| *Age* | −0.0074 | −0.0112 | −0.0062 |
| | (0.0172) | (0.0174) | (0.0172) |
| *Lev* | −0.0018** | −0.0018*** | −0.0018*** |
| | (0.0007) | (0.0007) | (0.0007) |
| *ROE* | 0.0093** | 0.0094** | 0.0096** |
| | (0.0045) | (0.0045) | (0.0045) |
| *SA* | −0.0872*** | −0.0687*** | −0.0861*** |
| | (0.0183) | (0.0189) | (0.0184) |
| *TobinQ* | 0.0153*** | 0.0153*** | 0.0157*** |
| | (0.0016) | (0.0016) | (0.0016) |
| *Size* | −0.0123*** | −0.0097*** | −0.0118*** |
| | (0.0025) | (0.0026) | (0.0025) |
| *Fixed* | −0.0804*** | −0.0834*** | −0.0816*** |
| | (0.0154) | (0.0153) | (0.0154) |
| *Growth* | 0.0044* | 0.0047* | 0.0046* |
| | (0.0025) | (0.0025) | (0.0025) |
| *Top*10 | 0.0362** | 0.0380*** | 0.0371*** |
| | (0.0143) | (0.0143) | (0.0143) |
| *Dual* | 0.0026 | 0.0031 | 0.0027 |
| | (0.0031) | (0.0031) | (0.0031) |
| *Cons* | −0.2003** | −0.1377 | −0.2031** |
| | (0.0873) | (0.0893) | (0.0876) |
| *Firm* | Yes | Yes | Yes |
| *Year* | Yes | Yes | Yes |
| N | 15,968 | 15,968 | 15,968 |
| adj. $R^2$ | 0.0429 | 0.0439 | 0.0427 |

mechanisms can effectively mitigate the misallocation of credit resources. These findings make two marginal contributions to the advancement of relevant theories. First, we expand the research boundaries of signaling theory. Existing studies have largely focused on the transmission mechanisms of genuine signals; this study extends the theory's application to the realm of false signaling in enterprises' AI technology disclosures and actual adoption, identifying the salient fact that corporate AI washing facilitates access to bank credit. It also confirms that external governance mechanisms, as a critical signaling context, impose a significant constraint on the efficacy of false signals. In doing so, this study enriches research on signaling theory in the context of false signal transmission and responds to the scholarly proposition by Connelly et al. (2011) that the signaling context plays a pivotal role in determining signal effectiveness. Second, we refine the theoretical framework underpinning the relationship between the digital economy and enterprises' access to bank loans. Against the backdrop of extant literature that predominantly emphasizes the positive impacts of the digital economy on corporate financing, this study uncovers the distorting effect of emerging opportunistic behaviors such as AI washing on credit resource allocation, as well as the channels through which external governance mechanisms inhibit this effect. This thus enriches the body of research on corporate financing in the digital economy era.

## Author agreement statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us. We understand that the Corresponding Author is the sole contact for the Editorial process. She is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

**Table 6**
Heterogeneity analysis results.

| Variables | (1) State-owned Enterprises | (2) Non-state-owned Enterprises | (3) High-tech Industries | (4) Non-high-tech Industries |
|---|---|---|---|---|
| *AIwashing* | 0.0088*** | 0.0041 | 0.0028 | 0.0110*** |
| | (0.0033) | (0.0031) | (0.0032) | (0.0032) |
| *Age* | −0.0035 | 0.0010 | −0.0420 | 0.0249 |
| | (0.0229) | (0.0267) | (0.0262) | (0.0224) |
| *Lev* | −0.0027*** | −0.0005 | −0.0008 | −0.0024*** |
| | (0.0008) | (0.0012) | (0.0011) | (0.0009) |
| *ROE* | 0.0067 | 0.0135** | 0.0066* | 0.0208*** |
| | (0.0057) | (0.0056) | (0.0035) | (0.0055) |
| *SA* | −0.0739*** | −0.1087*** | −0.0558** | −0.1127*** |
| | (0.0212) | (0.0324) | (0.0272) | (0.0238) |
| *TobinQ* | 0.0157*** | 0.0170*** | 0.0112*** | 0.0208*** |
| | (0.0028) | (0.0021) | (0.0021) | (0.0026) |
| *Size* | −0.0114*** | −0.0155*** | −0.0133*** | −0.0121*** |
| | (0.0035) | (0.0033) | (0.0043) | (0.0031) |
| *Fixed* | −0.0767*** | −0.0825*** | −0.0858*** | −0.0781*** |
| | (0.0206) | (0.0230) | (0.0224) | (0.0211) |
| *Growth* | −0.0018 | 0.0090** | 0.0059 | 0.0034 |
| | (0.0031) | (0.0037) | (0.0037) | (0.0035) |
| *Top*10 | 0.0060 | 0.0399** | 0.0259 | 0.0422** |
| | (0.0209) | (0.0184) | (0.0201) | (0.0186) |
| *Dual* | 0.0036 | 0.0012 | 0.0046 | −0.0011 |
| | (0.0047) | (0.0040) | (0.0046) | (0.0040) |
| *Cons* | −0.1451 | −0.2849** | 0.0236 | −0.3919*** |
| | (0.1112) | (0.1412) | (0.1256) | (0.1174) |
| *Firm* | Yes | Yes | Yes | Yes |
| *Year* | Yes | Yes | Yes | Yes |
| N | 7680 | 8288 | 6800 | 9168 |
| adj. $R^2$ | 0.0338 | 0.0562 | 0.0373 | 0.0487 |
| Fisher's Permutation test [P-value] | −0.005* | 0.008* | | |
| | [0.078] | [0.098] | | |

## CRediT authorship contribution statement

**Weiqi Liu:** Conceptualization. **Meifang Li:** Writing – original draft, Project administration, Methodology.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant numbers 72372093].

## Data availability

Data will be made available on request.

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